

WBS Task ID	Task Description	Dependencies (WBS Task ID)	MoSCoW Score	Impact Score	Effort Score	Modified MoSCoW-I/E Score	Duration (People-Days, Most Likely Time)	Scope Value
Tasks marked in red are out of scope and have been cut								
Graphics and Visual Assets								
g1	Placeholder testing graphics for player and world tiles		Must have	Low	Low	6	0.5	High
g2	Consistent visual assets (modern UI/pixel art game) for player	g1, gd1	Should have	High	Low	7	0.5	High
g3	Consistent visual assets (modern UI/pixel art game) for tiles (floor, elevation changes, hazards)	g1, w5	Should have	High	High	3	0.5	Low
g4	Consistent visual assets (modern UI/pixel art game) for world objects (doors, keys, switches)	g1, w8	Should have	High	High	3	0.5	Low
g5	Consistent visual assets (modern UI/pixel art game) for main menu	g1, ui3	Should have	High	High	3	0.5	Low
g6	Consistent visual assets (modern UI/pixel art game) for config menu	g1, ui6	Should have	High	High	3	0.5	Low
g7	Consistent visual assets (modern UI/pixel art game) for pause menu	g1, ui5	Should have	High	High	3	0.5	Low
g8	Consistent visual assets (modern UI/pixel art game) for interface elements	g1, ui7-9, ui17	Should have	High	High	3	0.5	Low
g9	UI elements with high-resolution options	g5, g6, g7, g8	Should have	High	High	3	0.5	Low
g10	Code block shape to differ by function	c1	Should have	High	High	3	0.5	Low
g11	Code block colour to differ by function	c1	Should have	High	High	3	0.5	Low
Audio and Sound Assets								
								No
a1	Small click sound when block placed	c1	Could have	Low	Low	4	0.25	Medium
a2	Snap sound when blocks connect	c1	Must have	High	Low	8	0.25	High
a3	Error sound when blocks cannot connect	c1	Must have	High	Low	8	0.25	High
a4	Footstep sound when character walks	gd2	Could have	Low	Low	4	0.25	Medium
a5	Jump sound when character jumps	gd3	Should have	High	Low	7	0.25	High
a6	Landing sound so movement feels realistic	gd3	Could have	Low	Low	4	0.25	Medium
a7	Success sound when code works	c1	Must have	High	Low	8	0.25	High
a8	Fail/error sound when code breaks	c1	Must have	High	Low	8	0.25	High
a9	Fail/error sound when action can't complete	c2	Must have	High	Low	8	0.25	High
a10	Completion sound when level finished	w12	Should have	High	Low	7	0.25	High
a11	Soft sounds for button clicks on menus	ui2	Should have	Low	Low	5	0.25	Medium
a12	Hover/select sound when choosing options	ui2	Should have	Low	Low	5	0.25	Medium
a13	Basic ambient sounds in the level	w12	Want	Low	Low	3	0.25	Low
a14	Level objects (doors, switches, moving platforms) to have simple sound cues	w8	Should have	Low	Low	5	0.25	Medium
a15	Sounds played when game starts up	ui3	Want	High	Low	5	0.25	Medium
a16	Sounds played when menu opens	ui2	Want	Low	Low	3	0.25	Low
User Interface								
ui1	Forced landscape screen orientation		Must have	High	Low	8	0.5	High
ui2	Base interface window class		Must have	Low	Low	6	0.5	High
ui3	Main menu (level select, quit)	ui2, lid6	Must have	High	Low	8	1	High
ui4	Configuration menu (accessibility, resolution, toggleable options)	ui3, ui5, lid7	Should have	High	High	3	3	Low
ui5	Pause menu (quit, config, return)	ui2, gd6, lid8	Must have	High	Low	8	1	High
ui6	Base interface element class/scene	w12	Must have	Low	Low	6	0.5	High
ui7	Interface element - time code executing for	ui6	Could have	High	High	2	0.5	Low
ui8	Interface element - time working on the level (toggleable option)	ui6, w12, sc3	Could have	High	High	2	0.5	Low
ui9	Interface element - blocks interacted with/changed (toggleable option)	ui6, sc4	Could have	High	High	2	0.5	Low
ui10	Code block interface objects	c1, lid10	Must have	High	Low	8	0.5	High
ui11	Code block menu, palette for selecting blocks	ui10, lid10	Should have	High	High	3	0.75	Low
ui12	Code block menu, workspace to snap them together	ui11, lid10	Should have	High	High	3	0.75	Low
ui13	Console view showing action output	c12, lid12	Could have	High	Low	6	0.75	High
ui14	Time control (pause/reset/step-by-step) buttons on interface	gd6	Could have	High	High	2	0.75	Low
ui15	Level select menu	ui3, lid9	Must have	High	Low	8	1	High
ui16	Dark mode functionality (UI/menu changing)	ui4	Must have	High	High	4	2	Medium
ui17	Interface element - start code execution button	ui6, gd4	Must have	High	Low	8	0.5	High
ui18	Tutorial Popup Window	ui2	Should have	Low	Low	5	1	Medium

User Experience								
ux1	Undo action feature	c19	Want	High	High	1	3	No
ux2	Redo action feature	c19	Want	High	High	1	3	No
ux3	Select block groups (long press)	c1	Should have	High	High	3	1	Low
ux4	Copy & paste block combination feature	c1	Could have	High	Low	6	2	High
ux5	Accessibility – colour blindness (palette changes)	ui4	Should have	Low	High	1	2	No
ux6	Accessibility – sound captions (poor hearing)	ui4	Should have	Low	High	1	4	No
ux7	Accessibility – text to speech (poor vision)	ui4	Should have	Low	High	1	4	No
ux8	Error messages on level/execution failure (consistent)	w12	Must have	High	Low	8	1	High
ux9	Error feedback is clear	ux8	Must have	High	Low	8	1	High
ux10	Swipe shortcuts for accessing different menus (world, blocks, pause, console)	gd8, ui5, ui11-ui13	Want	Low	High	-1	1	No
ux11	Intuitively understandable game world & elements	w12	Should have	High	High	3	1	Low
ux12	Intuitively understandable interface game menus & icons	ui3-ui5, ui15	Should have	High	High	3	1	Low
ux13	Intuitively understandable block palette & workspace	ui15, ui16	Should have	High	High	3	1	Low
ux14	Palette is collapsible on block workspace	ui11	Could have	High	Low	6	1	High
ux15	Console is collapsible or hideable/toggleable	ui13	Could have	High	Low	6	1	High
ux16	Automatic resolution adjustment based on device	ui1	Could have	High	High	2	3	Low
ux17	Scaling assets with resolution changes	ui1	Should have	High	High	3	2	Low
ux18	Minimal interface, fading/hideable objects	ui6	Could have	High	Low	6	2	High
ux19	User control over showing/hiding interfaces	ui6	Want	High	Low	5	2	Medium
ux20	Visual indicator for code blocks to link with each other (puzzle piece)	c1	Must have	High	High	4	0.5	Medium
ux21	Block highlighting on press/hold	ui10	Should have	High	Low	7	0.5	High
ux22	Automatic infinite loop detection	c23	Should have	High	Low	7	1	High
Game Design								
gd1	Player class		Must have	High	Low	8	0.5	High
gd2	Player can move in response to events (coding block calls)	gd8	Must have	High	Low	8	1	High
gd3	Player can jump	gd8	Should have	High	Low	7	1	High
gd4	Execute' function (button) to start code blocks running	c1	Must have	High	Low	8	0.5	High
gd5	Levels track starting state	w12	Should have	High	High	3	1	Low
gd6	Game execution can be paused	w12	Should have	High	High	3	1	Low
gd7	Game execution can be abandoned/reset during gameplay	gd5	Should have	High	High	3	1	Low
gd8	World environment for viewing the player	gd1	Must have	High	High	4	1	Medium
gd9	Player can track facing (relevant for coding blocks)	gd1	Could have	Low	Low	4	0.5	Medium
gd10	Step-by-step execution (ala debugger)	gd6	Want	High	High	1	4	No
gd11	World Z-Level Implementation	gd8	Should have	High	High	3	3	Low
gd12	Player can pick up world objects	gd1, w5	Must have	High	Low	8	2	High
gd13	Game camera	gd8	Must have	High	Low	8	2	High
gd14	Pannable camera during level pause/starting state	gd13, gd6	Must have	High	Low	8	1	High
gd15	Game camera following player during level execution	gd13, gd1	Must have	High	Low	8	1	High
Coding Blocks								
c1	Base coding block class		Must have	Low	High	2	3	Low
c2	Action block class	c1	Must have	Low	Low	6	2	High
c3	Movement action block	c2, gd2	Must have	High	Low	8	2	High
c4	Jump action block	c2, gd3, gd11	Should have	High	Low	7	1	High
c5	Action blocks can be chained/connected to each other	c2	Could have	High	High	2	1	Low
c6	If statement block class	c1	Must have	Low	Low	6	0.75	High
c7	If is safe block next (i.e. not hazard for facing or movement/jump action)	c6	Should have	High	High	3	2	Low
c8	If statements can be connected to each other	c6	Should have	High	High	3	1	Low
c9	While block class	c23	Must have	Low	High	2	1	Low
c10	While block to attach to if statement chains to repeatedly re-evaluate	c9	Should have	High	High	3	1	Low
c11	While break checks (counter keeps track of executions)	c10	Should have	High	High	3	1	Low
c12	Code blocks log to console interface when executing	c1	Could have	Low	Low	4	2	Medium

c13	Code action blocks log to console interface whether they were able to complete their action	c2, c12	Want	Low	High	-1	3	No
c14	Customisable block options on selection (i.e. select the break check, wait duration etc)	c11	Want	Low	High	-1	4	No
c15	Wait x second blocks	c2	Could have	Low	Low	4	0.75	Medium
c16	Function block class (block combinations)	c6	Want	High	High	1	5	No
c17	Function blocks auto-generated names	c16	Could have	High	High	2	1	Low
c18	Go-To X/Y movement action blocks (with different search algorithms)	c3	Want	Low	High	-1	7	No
c19	Action driven implementation of coding blocks (supporting undo/redo)	c1	Want	High	High	1	14	No
c20	Sensor check if conditional blocks	c6	Want	High	High	1	3	No
c21	Blocks can hold variables, customisable block options accommodate this	c14	Want	High	High	1	2	No
c22	Variable counter increment block	c21	Want	High	High	1	1	No
c23	For loop block class	c6	Must have	Low	High	2	2	Low
c24	Pick up and drop action blocks	c2, w13	Want	High	High	1	2	No
Level and Interface Design								
lid1	Level sketch for 1-1 (movement)		Should have	Low	Low	5	2	Medium
lid2	Level sketch for 1-2 (if statements, jumping)		Should have	Low	Low	5	2	Medium
lid3	Level sketch for 2-1 (for loops & long move orders)		Should have	Low	Low	5	2	Medium
lid4	Level sketch for 2-2 (while loops/conditionals & combined if statements, hazards)		Should have	Low	Low	5	2	Medium
lid5	Design sketch for gameplay with interface elements		Should have	Low	Low	5	3	Medium
lid6	Design sketch for main menu		Should have	Low	Low	5	2	Medium
lid7	Design sketch for configuration menu		Should have	Low	Low	5	2	Medium
lid8	Design sketch for pause menu		Should have	Low	Low	5	1	Medium
lid9	Design sketch for level select menu		Should have	Low	Low	5	1	Medium
lid10	Design sketch for block palette and workspace		Should have	Low	Low	5	2	Medium
lid11	Design sketch for score metrics/leaderboard		Could have	Low	Low	4	1	Medium
lid12	Level sketch for 3-1 (functions and collectables)		Could have	Low	Low	4	4	Medium
lid13	Level sketch for 3-2 (variables, sensors, and obstacles)		Could have	Low	Low	4	4	Medium
World and Level Implementation								
w1	Level 1-1 that introduces coding blocks (movement)	lid1, c3, w12	Must have	High	High	4	3	Medium
w2	Level 1-2 that introduces if statements and z levels (jumping)	lid2, gd3, gd11, w12	Should have	High	High	3	2	Low
w3	Level 2-1 that adds for loops for long movement chains	lid3, c6, w6, w12	Should have	High	High	3	2	Low
w4	Level 2-2 that introduces hazards with looping ifs (while next block valid/safe move forward)	lid4, c9, w12	Should have	High	High	3	3	Low
w5	Level/world tile setup	gd1, gd8	Must have	High	High	4	1	Medium
w6	Level hazard tiles (water/pit/fire)	w5	Should have	High	High	3	0.75	Low
w7	Level door tiles that block progress but are passable/open with a key	w5	Want	Low	Low	3	1	Low
w8	Level object class (collectable that responds to player contact)	w5	Want	Low	Low	3	1	Low
w9	Level key objects	w7, w13	Want	Low	Low	3	0.5	Low
w10	Level switch objects	w8	Could have	Low	Low	4	0.5	Medium
w11	Level moving platform objects	w8	Want	Low	High	-1	0.5	No
w12	Base level class and scene	w5	Must have	Low	High	2	1	Low
w13	Collectable/holdable world objects	w8	Could have	Low	Low	4	1	Medium
w14	Score influencing collectable objects	w13	Could have	High	Low	6	0.5	High
w15	Path blocking world objects	w8	Could have	Low	Low	4	0.75	Medium
w16	Level 3-1 that introduces functions and collectables	lid12, c16, w12-w14	Could have	Low	Low	4	3	Medium
w17	Level 3-2 that introduces sensors, variables, and obstacles	lid13, c20, c22, w12, w15	Could have	Low	Low	4	3	Medium
Tutorialisation								
t1	Tutorial popup - code block snapping	ui18, c1	Must have	High	High	4	1	Medium
t2	Tutorial popup- action block execution & moving	ui18, c3	Should have	High	High	3	1	Low
t3	Tutorial popup- elevation and jumping	ui18, c4, gd3, gd11	Should have	High	High	3	1	Low
t4	Tutorial popup- hazards	ui18, w6	Should have	Low	High	1	1	No
t5	Tutorial popup- if statements	ui18, c6	Should have	High	High	3	1	Low
t6	Tutorial popup- while statements	ui18, c9	Should have	High	High	3	1	Low
t7	Long press hints for concepts and UI elements	ui18, ui2	Could have	High	High	2	2	Low

t8	Tutorial popup- for loops	ui18, c23	Could have	High	High	2	1	Low
t9	Tutorial popup - collectables	ui18, w8	Could have	High	High	2	1	Low
t10	Tutorial popup - variables	ui18, c22	Could have	High	High	2	1	Low
t11	Tutorial popup - functions	ui18, c16	Could have	High	High	2	1	Low
t12	Tutorial popup - sensors	ui18, c20	Could have	High	High	2	1	Low
Social Connectivity								
sc1	Local metric/progress storage	w12, lid10	Must have	High	Low	8	1	High
sc2	Metric recorded - levels completed	sc1	Could have	High	Low	6	0.5	High
sc3	Metric recorded - time taken per level	sc1	Could have	High	Low	6	0.5	High
sc4	Metric recorded - block actions taken per level	sc1	Could have	High	Low	6	0.5	High
sc5	Badge system	sc2	Could have	High	Low	6	2	High
sc6	Visible achievements/metrics	sc2	Should have	High	Low	7	2	High
sc7	Custom generated usernames for users		Want	Low	Low	3	1	Low
sc8	Server-side leaderboard system API	sc1	Could have	High	High	2	7	Low
sc9	User authentication and security	sc7, sc8	Must have	High	High	4	3	Medium
sc10	Leaderboard system – online sync	sc8	Could have	High	High	2	7	Low
sc11	Metric recorded - collectables attained	sc1, w14	Could have	High	Low	6	1	High
Testing and Playtesting								
p1	Ethics application for open-day user testing		Could have	High	High	2	28	Low
p2	Stakeholder testing demo of playable level (repeated task)	w1	Should have	High	Low	7	1	High
p3	Testing class		Must have	Low	Low	6	5	High
p4	Open-Day user testing	p1	Could have	High	High	2	2	Low
p5	Action block behaviour and logging integration testing	p3, c3, c12, gd2	Should have	Low	High	1	3	No
p6	Tutorialisation trigger integration testing	p3, ui18	Could have	Low	High	0	3	No
p7	Level load and reset (setup/spawning) test	p3, w12	Should have	Low	High	1	2	No
p8	Sensor block behaviour unit test	p3, c20	Could have	Low	High	0	1	No
p9	Execution of block command unit test	p3, gd4	Should have	Low	High	1	3	No
p10	Configuration menu option unit tests	p3, ux5-ux7	Should have	Low	High	1	7	No